

HOSPITAL ANALYTICS SYSTEM

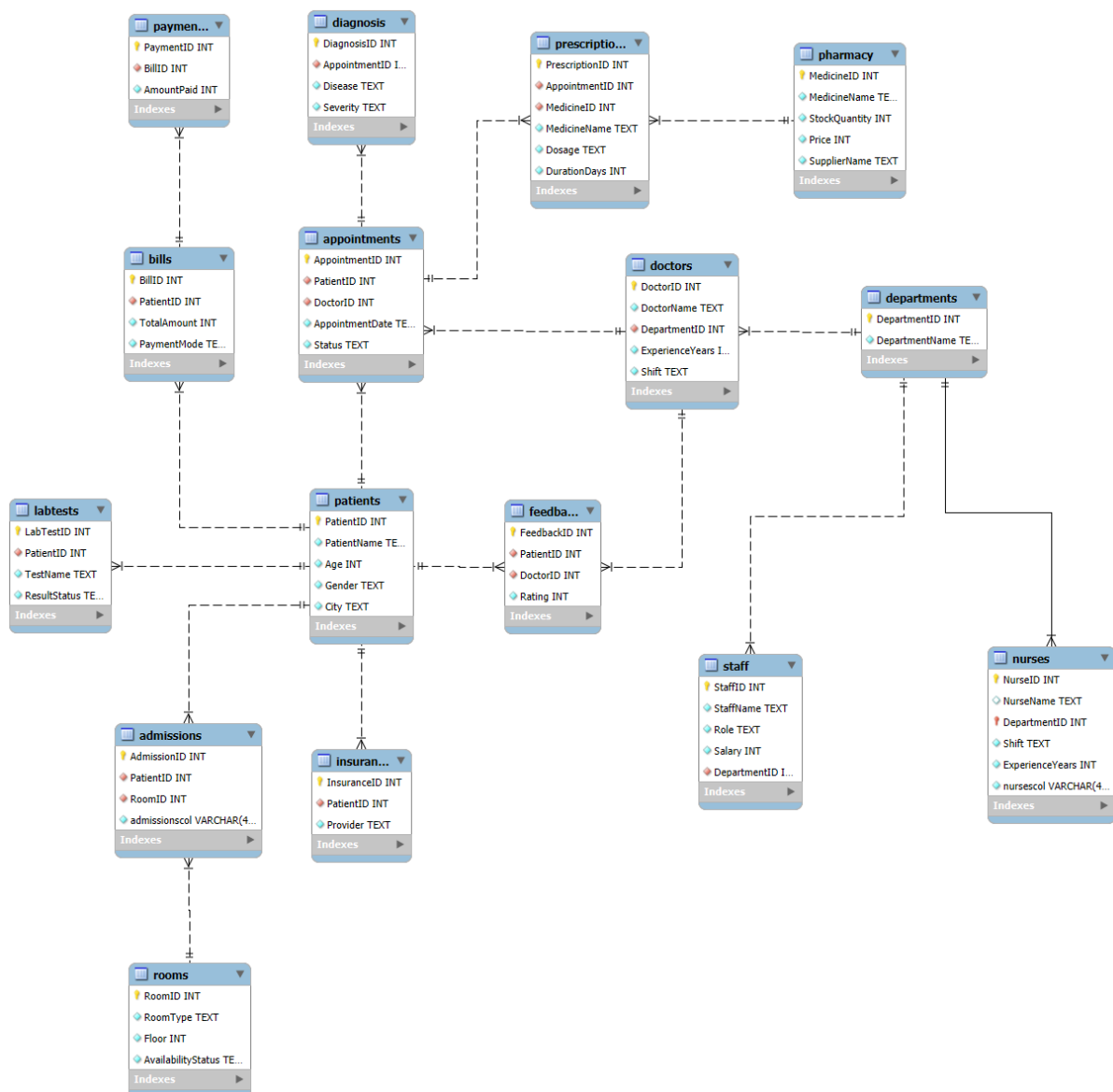
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GitHub URL: https://github.com/sivasks2004/Hospital_Analytics_System.git

Table Schema:



BASIC LEVEL

- Write a query to find all patients whose age is above 65.**
- Retrieve all appointments scheduled for today.**
- Find the total number of patients available in each city.**
- Display all doctors working in the Cardiology department.**
- Find the total number of appointments handled by each doctor.**
- Retrieve patients who do not have insurance records.**
- Find all bills where the total amount exceeds ₹50,000.**
- Display the top 10 highest payment amounts.**
- Find the total number of rooms available floor-wise.**
- Retrieve all appointments with status = 'Cancelled'.**
- Find the average patient age grouped by gender.**
- Retrieve doctors having more than 10 years of experience.**
- Find the total number of lab tests conducted for each patient.**
- Display all medicines with stock quantity below 100.**
- Find all patients who visited more than once.**
- Retrieve departments having more than 20 doctors.**
- Find all feedback ratings below 3.**
- Display the latest 20 admission records.[Admission Date column missing]**
- Find the total hospital revenue generated through bills.**
- Retrieve the count of appointments month-wise.**

INTERMEDIATE LEVEL

- Find the top 5 doctors who handled the highest number of appointments.**
- Retrieve patients who have appointments in multiple departments.**
- Find the department generating the maximum revenue.**
- Display patients who underwent more than 3 lab tests.**
- Find the most prescribed medicine in the hospital.**
- Retrieve doctors whose average feedback rating is above 4.5.**
- Retrieve the second highest billed patient.**
- Find doctors whose appointments are mostly cancelled.**
- Retrieve the average bill amount generated by each department.**
- Find the top 10 cities generating maximum hospital revenue.**
- Display medicines that are prescribed by more than 5 doctors.**
- Find patients who never provided feedback after appointments.**
- Retrieve all duplicate patient records based on name and phone number.**
- Find the percentage contribution of each doctor to total hospital revenue.**
- Find the percentage contribution of each department to total hospital revenue.**
- Display doctor performance rank using: Total appointments, Revenue generated, Average feedback rating**

ADVANCED SQL (STORED PROCEDURE, TRIGGERS)

- Write a query to identify suspicious payments where payment amount exceeds bill amount.
- Find patients who revisit the hospital within 7 days for the same disease.
- Detect duplicate insurance claims for the same patient and diagnosis.

-- Payment status automation

delimiter //

create trigger paymentAutomation

after insert on payments

for each row

begin

declare totalAmt int;

declare paidAmt int;

select totalAmount into totalAmt

from bills where billID=new.BillID;

select coalesce(sum(amountPaid),0) into paidAmt

from payments where billID=new.BillID;

if TotalAmt = paidamt then

 update bills set status="Paid" where billId=new.BillID;

elseif TotalAmt<paidAmt then

 update bills set status="suspicious" where billId=new.BillID;

else

 update bills set status="Partially paid" where billId=new.BillID;

end if;

end //

delimiter ;

-- Duplicate insurance prevention

drop procedure if exists DIP;

delimiter //

create procedure DIP

(IN p_InsuranceID int,

IN p_PatientID int,

IN p_Disease text,

IN p_Provider text)

begin

declare CountInsurance int;

select count(i.insuranceID) into CountInsurance

from insurance i inner join appointments a on i.patientID=a.patientID

inner join diagnosis d on a.appointmentID=d.appointmentID

where i.patientID=p_PatientID and d.disease=p_Disease;

if countInsurance>1 then

signal SQLState '45000'

SET MESSAGE_TEXT = 'Prohibited: Duplicate Entry';

else

insert into insurance(insuranceID,patientID,Provider)
values(p_InsuranceID,p_PatientID,p_Provider);

end if;

end//

delimiter ;

call DIP(1,3597,'Allergy','LIC');

-- show procedure status like 'DIP';

-- Room Booking

drop procedure if exists ARB;

delimiter //

create procedure RB(in p_roomID int)

begin

declare rStatus text;

declare rID int;

select roomID, AvailabilityStatus into rID, rStatus from rooms where roomID=p_roomID;

if rID is null then

 signal sqlstate '45000'

 set MESSAGE_TEXT ='Room Not Available';

elseif rStatus='Available' then

 update rooms set AvailabilityStatus='Occupied' where roomID=p_roomID;

else

 signal SQLState '45000'

 SET MESSAGE_TEXT = 'Prohibited: Duplicate Entry';

end if;

end //

delimiter ;